SLEEPCL / RESEARCH SOFTWARE

Sleep staging, slow oscillations & timed click decisions

An illustrated walkthrough of the signal processing, model choices, development decisions and selected results.

What was built

SleepCL combines EEG-based classification of W, N1, N2, N3 and REM with slow-oscillation (SO) detection and a rule-based scheduler for auditory-click timing. Recorded Sleep-EDF EEG is replayed sequentially to explore how model confidence and SO phase jointly determine click decisions.

FEATURE EPOCHS	VALIDATION	REPLAY
8,309	**4 folds**	**58**
five participants	held-out participant each fold	logged clicks in 180 seconds

The two connected workflows

OFFLINE ANALYSIS AND MODEL DEVELOPMENT	STREAMED REPLAY AND CONTROLLER
Scored 30-second epochs → seven features → candidate classifiers → participant-separated validation	EEG chunks → rolling 30-second predictions → N2/N3 and confidence gate → SO candidate → click scheduler

Questions answered in this report

What is timed, which signals are used, how the sleep stages are estimated, how online-style decisions differ from offline analysis, how the click target and timing were determined, what went wrong during development, and what the figures actually show.

Scope in one sentence

The proposed stimulus is a brief sound during eligible NREM sleep; the present implementation records click decisions for analysis rather than driving audio hardware.

STUDY IDEA

What is targeted, and where?

A short answer to the stimulation question before the technical details.

Stimulus and recording locations

The intervention concept is a brief auditory click intended to coincide with a selected portion of a slow oscillation during N2 or N3 sleep. Fpz–Cz is the primary scalp EEG derivation used for staging and SO tracking. Pz–Oz was later added as an optional second signal for checking the sleep-stage decision. The electrodes measure activity; they are not stimulation electrodes.

Why the phase matters

Slow oscillations alternate between down- and up-state activity. Auditory stimulation timed near the up-state is the scientific motivation: timing matters because a click at a different part of the cycle may have a different effect. Here, the target is operationalised as a numerical phase of the filtered EEG, with a tolerance window around it.

Two offline references — different purposes

QUESTION	AVAILABLE REFERENCE	WHAT IS COMPARED
Is the predicted sleep stage right?	The Sleep-EDF hypnogram: stages scored offline from the recording	Streamed or batch model predictions versus scored W/N1/N2/N3/REM labels
Did a scheduled click fall near the intended SO phase?	A retrospective Hilbert phase calculated from a longer SO-filtered segment	The streaming estimate / click times versus a separate offline phase calculation

Useful terminology

“Annotated” means the sleep stage supplied by the offline-scored hypnogram. It is the reference used for evaluation; it is not another prediction made by the real-time classifier.

STAGING / METHODS

How the sleep-stage classifier sees the EEG

Input, preprocessing and the seven numbers extracted from each 30-second window.

Preparation of labelled epochs

The analysis used nine Sleep-EDF nights from five people. The hypnogram supplies 30-second W, N1, N2, N3 and REM labels (original stages 3 and 4 are combined into N3). At 100 Hz sampling, the nominal 50 Hz notch lies at Nyquist and is skipped; a causal 0.3–35 Hz band-pass filters the EEG. The signal is divided into 30-second intervals matched to the labels. A valid-in-bed selection excludes large parts of out-of-bed Wake; this is useful for sleep-focused development but leaves comparatively few Wake examples.

Five spectral features + two time-domain features

FEATURE	CALCULATION / ROLE
Delta, 0.5–4 Hz	Slow activity, typically more prominent in deep NREM sleep
Theta, 4–8 Hz	Low-frequency spectral content
Alpha, 8–12 Hz	Activity that can help distinguish Wake from NREM in some recordings
Sigma, 12–16 Hz	Spindle-range activity relevant to N2
Beta, 16–30 Hz	Higher-frequency activity
Standard deviation (µV)	How much the EEG amplitude varies in an epoch
Mean absolute successive difference (µV)	A simple measure of sample-to-sample change

Welch's method estimates power spectral density using 4-second segments with 50% overlap. Each bandpower is divided by total 0.5–30 Hz power and converted to log10 scale; the time-domain statistics use µV. Each labelled epoch therefore becomes a seven-value input vector.

Why these features?

They describe interpretable characteristics of sleep EEG and are inexpensive to recompute every few seconds during replay. They are not intended to capture every possible marker used by an expert scorer.

STAGING / MODELS

Why compare four staging models?

The same seven features were given to models of increasing flexibility.

MODEL	HOW IT WORKS	WHY IT WAS INCLUDED
Logistic regression (LR)	Linear decision boundaries; feature standardisation	Fast, interpretable starting baseline
Linear SVM	Margin-based linear classifier; calibrated probabilities	Alternative linear boundary
Random forest (RF)	500 decision trees voting across feature splits	Nonlinear interactions without scaling
Multilayer perceptron (MLP)	Small neural network: 7 → 64 → 32 → 5 classes; ReLU, scaled features	Tests nonlinear combinations without a raw-EEG CNN

What the MLP adds

An MLP is a feed-forward neural network. Its two hidden layers combine the seven EEG features nonlinearly, allowing interactions that a linear LR or SVM cannot represent directly. This made it a sensible intermediate step: more expressive than a linear model, but much smaller and simpler than training a convolutional network on raw EEG.

How the comparison was run

Four development folds each trained on three participants and validated on a fourth among SC400–SC403; SC404 was reserved from all four. Each fold used the same feature definition and training settings. The results are an across-participant development comparison, not a final estimate from an untouched independent cohort.

Beyond the classical models

An EEGNet-like compact convolutional neural network was explored to learn features directly from EEG windows. The architecture and a separate inference path exist, but its training and evaluation were not completed to a comparable standard, so no head-to-head performance claim is made here.

STAGING / RESULTS

What the model comparison shows

Four validation participants, four classifiers and three complementary performance metrics.

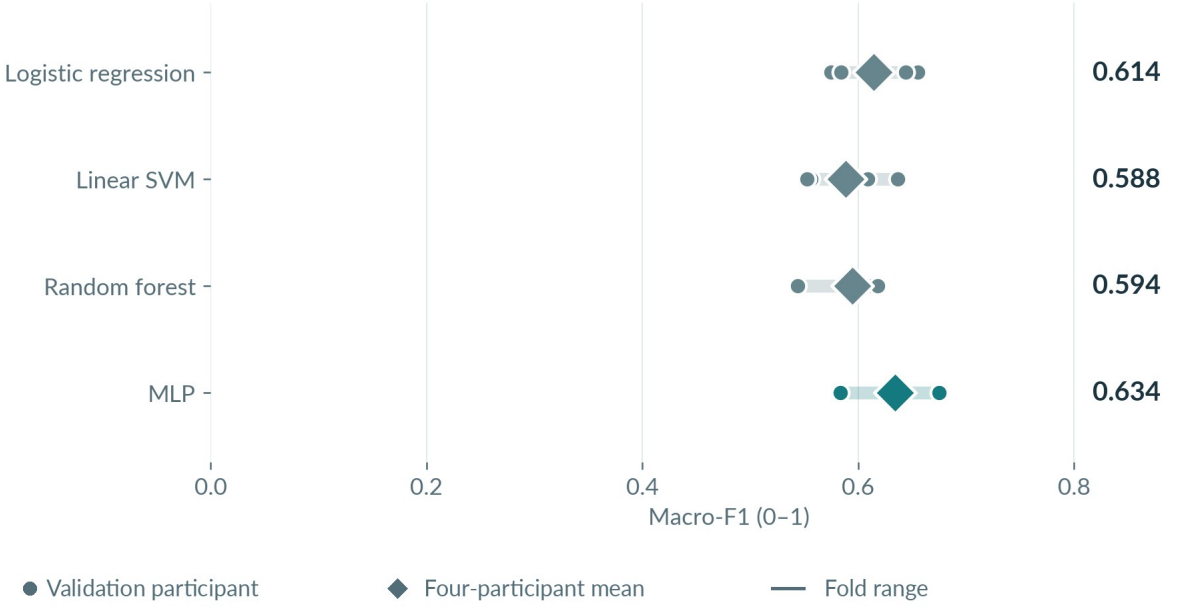


Figure. Each dot is one held-out participant; the diamond is the mean across the four participants and the horizontal line spans the observed fold values. Macro-F1 gives equal weight to the five sleep stages.

BEST MEAN ACCURACY	BEST MEAN BALANCED ACC.	BEST MEAN MACRO-F1
79.2%	66.8%	0.634
MLP	MLP	MLP

How to interpret the scores

Accuracy is the share of all epochs classified correctly. Balanced accuracy averages the recall of the five stages, so large classes do not dominate. Macro-F1 averages each stage's F1 score (a balance of precision and recall), again giving uncommon stages equal influence.

The MLP is highest on all three four-fold means, but LR is close, and model ranks vary across participants. These results support comparing their behaviour in the stimulation gate; they do not establish a single universally best stager.

STAGING / INTERPRETATION

What the stage confusion matrix means

Percentages are conditional on the reference stage, not percentages of the whole dataset.

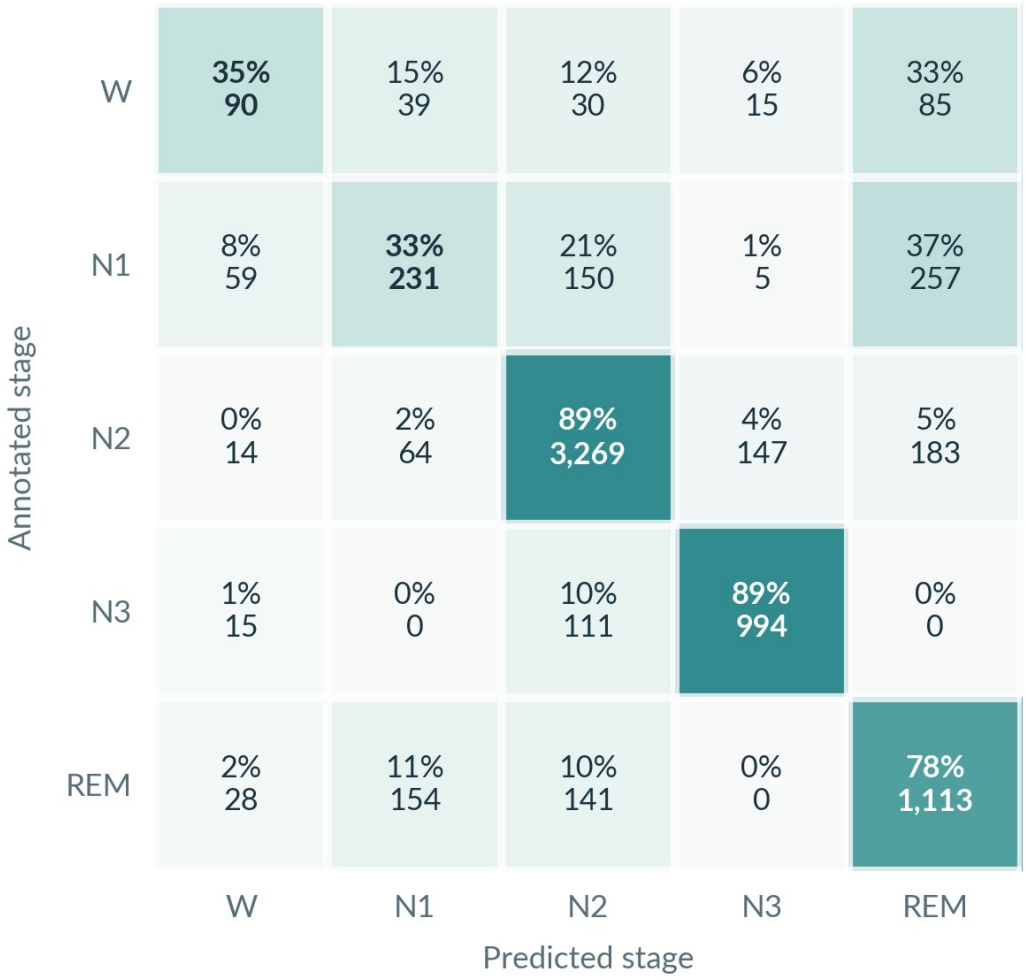


Figure. Pooled MLP validation matrix: rows = offline-scored ("annotated") stages; columns = model predictions. Each cell shows the percentage of that row assigned to a predicted class and the corresponding epoch count.

Reading one row

For example, 259 epochs were annotated Wake. The MLP labelled 90 as Wake (34.7% recall); the remaining 169 were called N1, N2, N3 or REM. The diagonal cell in each row therefore gives recall: the fraction of that actual stage recognised correctly.

WAKE RECALL	N1 RECALL	N2 / N3
34.7%	**32.9%**	**88.9 / 88.8%**
90 / 259	231 / 702	stronger recall

Because Wake and N1 are harder to recognise, even a model with good overall accuracy can misclassify Wake as N2/N3. That is consequential when the prediction controls whether a click is allowed.

DEVELOPMENT / FAILURE MODE

The unexpected Wake-to-NREM problem

A key development finding that changed the controller design.

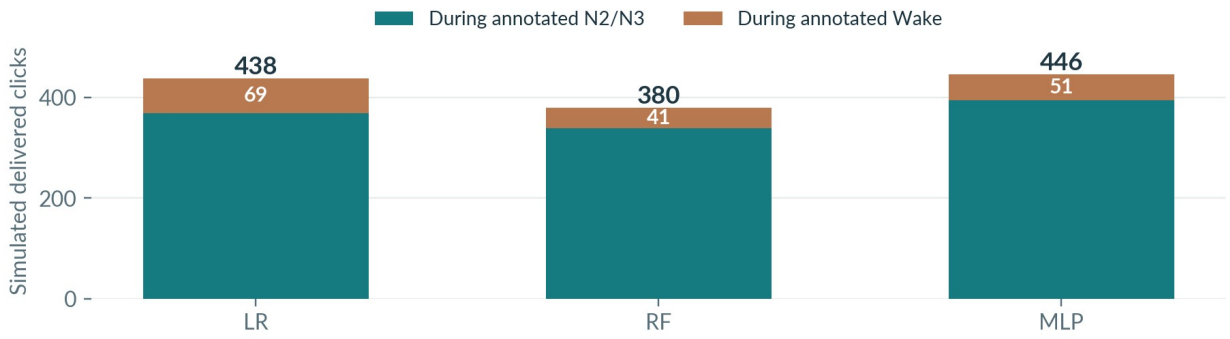


Figure. Initial single-channel stress tests: teal segments count logged clicks during annotated N2/N3, and orange segments count clicks during annotated Wake in the sampled test windows.

What triggered the investigation

Some Wake-labelled windows were repeatedly classified as N2 or N3 with apparently strong confidence. Increasing the probability threshold alone could not reliably reject such confident mistakes. The issue was particularly visible when Fpz–Cz showed strong low-frequency activity that resembled deep sleep.

In these selected one-channel tests, Wake clicks were counted for LR (69), RF (41) and MLP (51). These are counts from earlier exploratory windows, not rates or results from the later four-fold validation study.

What the second EEG channel was meant to contribute

Inspecting Pz–Oz suggested a different spectral picture: relatively more posterior higher-frequency content in problematic Wake windows. The design hypothesis was that agreement between frontal and posterior evidence would reject some false NREM decisions.

DEVELOPMENT / DESIGN CHANGE

Two-channel agreement gating

A conservative decision rule, not a new jointly trained classical classifier.



Figure. Fpz–Cz and Pz–Oz go through parallel preprocessing, seven-feature extraction and separate uses of the same classifier. Both results must pass before the SO controller may operate.

Precisely what the code does

The classical runner applies its single-channel, Fpz–Cz-trained classifier separately to both channels. For each, it checks that the most likely stage is N2 or N3, that its probability exceeds the chosen threshold, and that the top two class scores are sufficiently separated. Both channels must pass; one may say N2 and the other N3. The SO phase itself still comes from Fpz–Cz.

What did the first tests suggest?

In one exploratory RF example with this two-channel rule, no clicks occurred during the Wake-labelled portion and the mean logged event rate was about 2.02 per minute. That result is encouraging, but it was obtained in a particular set of windows rather than a full identical-window before/after comparison. It shows why the approach was worth exploring, not a guarantee that it eliminates Wake events.

An important implementation nuance

The model was trained on Fpz–Cz features and then also applied to Pz–Oz features. This is a practical agreement check; it does not replace a future, properly trained two-channel model.

STREAMING / METHODS

Offline evaluation vs streaming predictions

Recorded EEG serves two roles: a labelled dataset for training and an ordered signal for online-style replay.

Offline model development

In the batch workflow, each 30-second EEG epoch is extracted from the recording, converted to features and paired with its scored hypnogram stage. Classifier predictions are compared to these reference labels to calculate accuracy, class recall and confusion matrices.

Online-style replay

In replay, EEG arrives in simulated 50-ms chunks. A causal filter and ring buffer accumulate the incoming samples. The classifier uses the most recent 30-second buffer, but refreshes stage probabilities only every 5 seconds; averages of the last three probability vectors reduce flicker. SO candidate detection runs at the faster incoming-sample time scale.

Why the timing and filtering details matter

A staged decision can only use samples that have arrived by its timestamp. Training uses fixed epochs cut from longer filtered EEG; replay uses rolling windows and restarts its filter state at the chosen segment start. These are close but not identical input contexts, particularly during the initial warm-up period.

Top-class probability is the score assigned to the most likely stage. Top-two margin is the difference between the highest and second-highest scores: if N2 has 0.76 and N3 has 0.21, the margin is 0.55. At min_p=0.70 and min_margin=0.25, both the top probability and the margin must pass.

Answer to “did you compare offline and online?”

Yes, with a distinction: the model was evaluated offline against the scored hypnogram, and its streamed predictions were compared against those same labels during replay. Separately, SO click phase was assessed against a retrospective full-segment phase estimate. Neither offline computation should be called an independent physiological ground truth.

STREAMING / RESULTS

Reading one streamed staging example

The example uses a prespecified LR model and 180-second NREM recording interval.

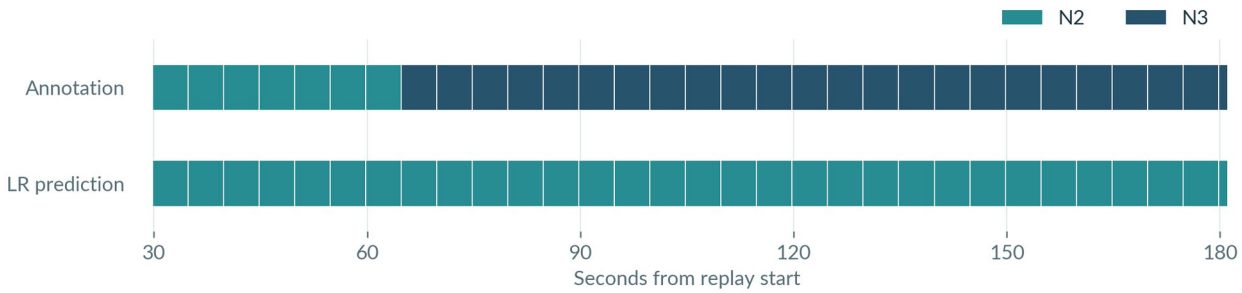


Figure. Upper strip: reference hypnogram stages at each 5-second update. Lower strip: the classifier's prediction. All 31 predictions were N2, while the reference was N2 for seven and N3 for 24 updates.

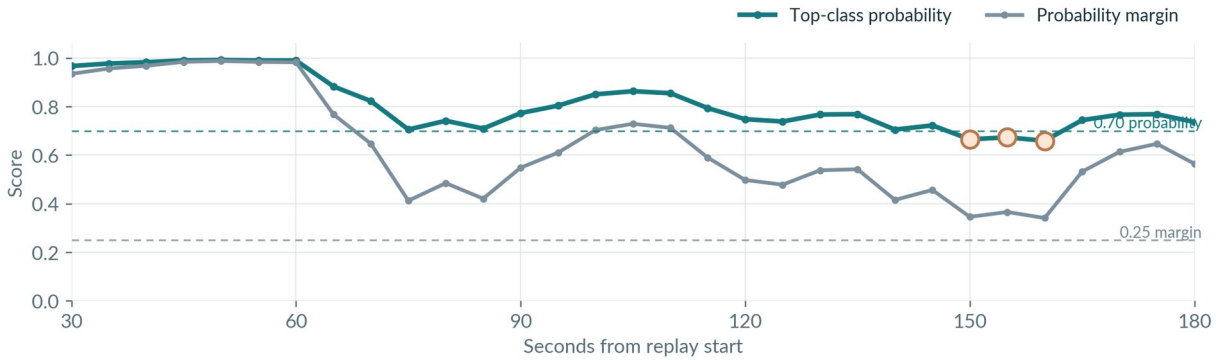


Figure. The upper line is the model's most likely class probability and the lower line is the gap to its runner-up class. Dashed horizontal lines show the 0.70 probability and 0.25 margin thresholds used for gating.

Why show LR if the MLP scored better on average?

LR was the original simple baseline and the preselected, compatible Fold-1 model for this particular engineering replay. The MLP ranked highest in mean development-validation metrics; it was not thereby established as the final closed-loop model.

The 7/31 exact-stage agreement is low because the model called N3 epochs N2. For this specific gate, either N2 or N3 was allowed, provided confidence criteria also passed. Three staging updates had a top-class probability below 0.70.

SLOW OSCILLATIONS / METHODS

How the SO target is detected

A continuously updated 12-second history is used to estimate the current oscillation phase.

STEP	WHAT IS CALCULATED	WHY IT IS NEEDED
1. Slow-band filter	Fpz–Cz with causal 0.3–1.2 Hz IIR band-pass	Emphasise slow oscillations
2. Trailing window	Last 12 seconds of SO-filtered samples	Provide signal history for a phase estimate
3. Hilbert phase	Analytic phase at the newest sample in that window	Estimate where the signal is within an oscillation
4. Eligibility	Circular phase distance ≤0.25 rad and robust amplitude ≥40 µV	Require a near-target phase and a sufficiently strong pattern
5. Edge detection	Register a candidate only on entering eligibility	Avoid repeatedly counting the same eligible region

What do the thresholds mean?

The target phase is an angle on a −π-to-π cycle; tolerance ±0.25 rad is about ±14°. “Robust amplitude” is the 95th minus the 5th percentile of the last 12 seconds, measured in µV. The phase and amplitude tests must both hold. “Eligible region” is simply the set of moments when both conditions pass.

Does the detector wait 12 seconds between decisions?

No. It needs 12 seconds of history before the first estimate, but then updates on each incoming sample. Edge detection issues a new candidate when the accepted phase/amplitude state changes from false to true, not once every 12 seconds.

Why use phase rather than only voltage peaks?

The goal is to schedule events around a defined part of the SO cycle. A phase estimate supports an angular tolerance window; it also makes timing discrepancies measurable later. It is not the same algorithm as simply finding local maxima in the filtered trace.

SLOW OSCILLATIONS / CALIBRATION

Why phase calibration became necessary

Nominal online phase and offline-assessed phase at click time did not coincide consistently.

The original discrepancy

The early target was 0 rad, close to a positive reference peak under the chosen signal convention. But when phases were recalculated retrospectively at the logged click times, the errors had both an offset and a broad spread. Possible contributors included the phase shift of causal filtering, the endpoint of the trailing Hilbert window, sample-level scheduling and any intended delivery delay. The tests did not isolate one cause.

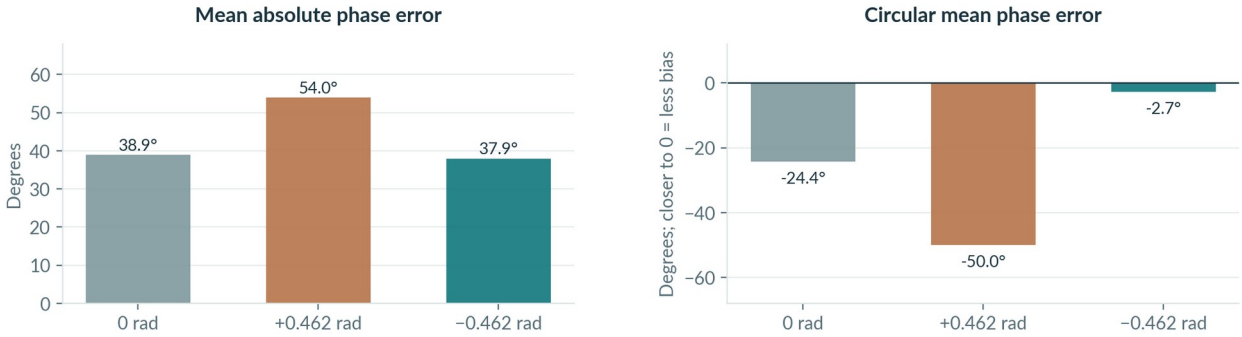


Figure. Exploratory comparison of three target settings. The left panel shows mean absolute retrospective phase error; the right panel shows signed circular mean error (the systematic offset). Values nearer zero are better, but the two measures answer different questions.

What was learned from the sign test

A TRAIN-sample circular mean error of −0.462 rad prompted a proposed +0.462 rad target correction. In the evaluated subset, that made the absolute error and bias worse. Applying −0.462 rad instead reduced the signed bias from about −24° to −3°, while the mean absolute error changed only slightly (38.9° to 37.9°).

Later simulations used −0.4919 rad as an operating target. The essential lesson is that the chosen phase convention and the timing at which phase is evaluated matter; reducing average bias is different from producing tightly distributed phase errors.

CONTROLLER / METHODS

How the controller schedules a click

A finite-state “armed” scheduler converts suitable SO candidates into timed events.

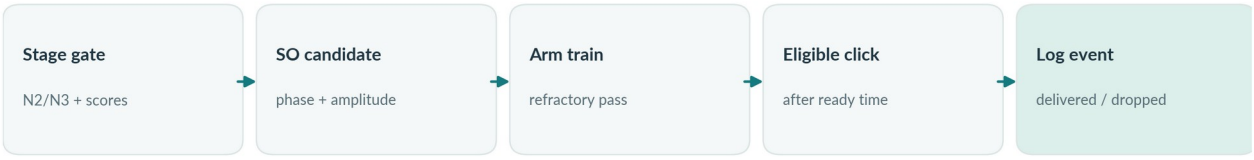


Figure. The full logic combines a permitted sleep state, an SO candidate, an armed train and an eligible click at or after its earliest allowed time before recording a delivered event.

What is a train? What does “armed” mean?

A train is a planned sequence of one or two auditory clicks. An eligible SO candidate can start (“arm”) a train after staging, confidence and refractory checks. With zero configured waiting delay, that same candidate may also deliver the first click; otherwise the controller waits for a later candidate before its deadline. For a second click, it waits for the minimum inter-click interval and another eligible candidate. Conditions are rechecked at click time.

CONTROL	EXAMPLE SETTING	EFFECT
Allowed stage and confidence	N2/N3; p ≥0.70; margin ≥0.25	Stage estimate must pass
Clicks per train	1 in displayed replay; optional 2	One event or a possible second event
Train refractory	1.0 second	Limit how quickly new trains start
Minimum inter-click interval	1.0 second	For a second click, wait at least this long
Armed maximum wait	2.0 seconds	Stop waiting when no eligible candidate arrives
Simulated latency parameter	0.0 seconds in displayed replay	Earliest permitted wait before looking for a later candidate

About hardware delay

A real audio system has acquisition, processing, driver and speaker delays that would have to be measured and compensated. The current click_latency_sec parameter is a scheduler waiting constraint, not a measurement or model of that hardware onset delay.

CONTROLLER / RESULTS

What the 180-second SO waveform shows

The full recording excerpt gives the overall number and distribution of eligible event opportunities.

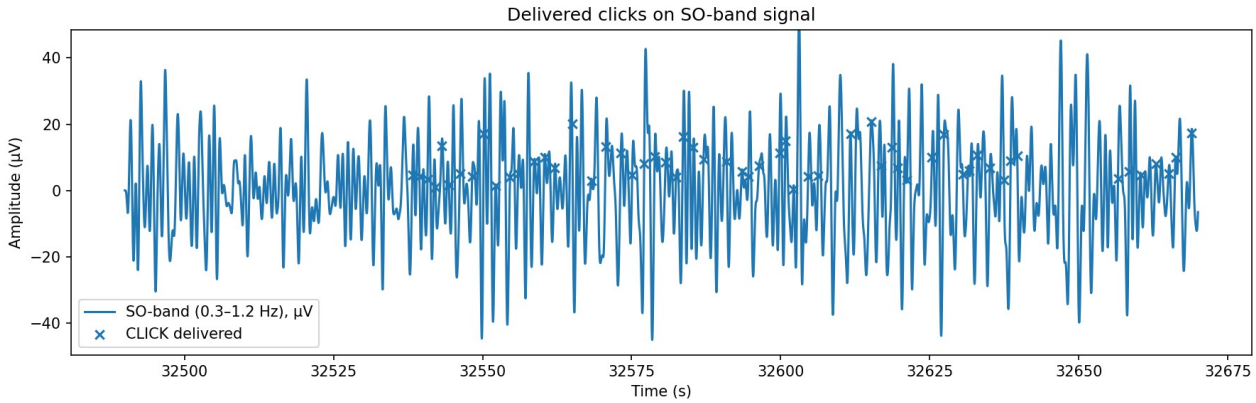


Figure. Blue line: Fpz–Cz after 0.3–1.2 Hz filtering. Crosses: logged click timestamps from a 180-second simulation. The full view is useful for event density, not for resolving individual cycles.

SO CANDIDATES	COMPLETED TRAINS	LOGGED CLICKS
110	58	58
eligibility onsets	one click per train	19.33 per minute

What happened, and how to read it

The detector found 110 moments when its phase and amplitude conditions first became eligible. The scheduler logged 58 of these as clicks after applying current stage, confidence and timing rules. The other 52 candidates were opportunities rather than failed sound deliveries; the candidate total alone does not identify which rule prevented each one from becoming a click.

Among the 58 events, eight occurred during reference N2 and 50 during reference N3. This chosen interval contained only N2/N3, so it illustrates within-NREM behaviour, not the ability to withhold clicks during Wake or REM.

CONTROLLER / SIGNAL VIEW

Individual oscillations: a 20-second close-up

The same replay, now with enough horizontal resolution to see event timing on the waveform.

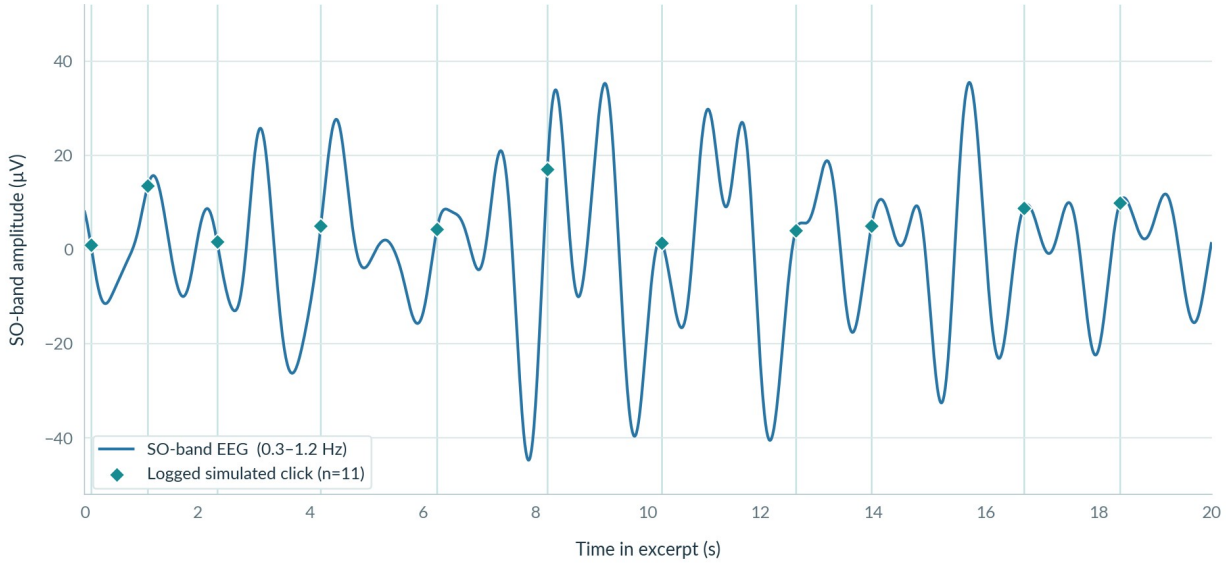


Figure. The 20-second interval includes 11 logged click events (small teal markers). The line shows SO-band EEG, and time is relative to the beginning of this excerpt.

What do the markers mean?

The clicks do not all sit exactly at the visible waveform maxima. The detector uses Hilbert phase at the end of a moving window, rather than directly identifying positive peaks on the plotted trace. It also accepts a band of phases (±0.25 rad), so timestamps near a peak are not expected to coincide at one exact sample.

For each marker, an SO candidate was accepted and the controller's sleep-stage, probability, margin and scheduler conditions were met. Comparing the marked locations with the trace helps explain the mechanism; it should not be read as a plot of the exact physical arrival time of a sound.

Why show both waveform views?

The 180-second figure answers "how many and how often?" The zoom answers "where do the markers fall relative to the shape of individual oscillations?"

CONTROLLER / INTERPRETATION

Reading the click-time and phase figure

Separate plots address sleep-state placement and numerical phase placement.

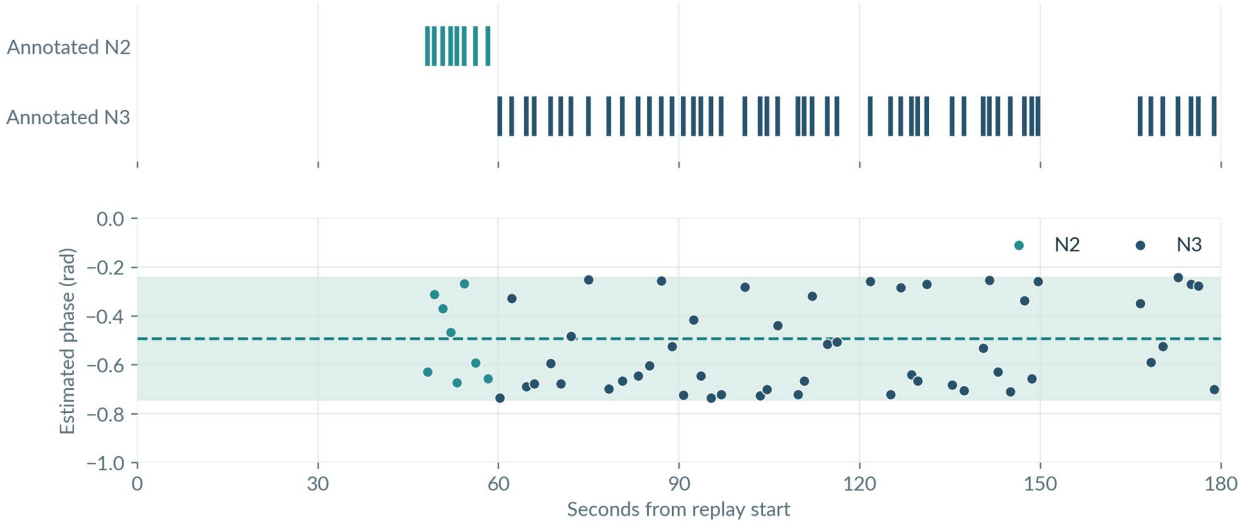


Figure. Upper panel: times of the 58 logged clicks, grouped by offline-scored N2 versus N3. Lower panel: the estimator's phase at each click; the shaded horizontal band is the accepted target range of −0.4919 ±0.25 rad.

How to interpret each panel

In the upper panel, each vertical mark is one logged event. Its row is determined by the hypnogram label at that moment (eight in N2, 50 in N3). This is why the plot can answer "which sleep stages contained the clicks?" even though the live classifier predicted N2 throughout.

In the lower panel, each point is the phase estimated by the streaming detector when a click was logged. The shaded band is the configured tolerance. All 58 points lie within it because the phase condition is part of the decision rule—not because an independent offline phase analysis confirmed every event.

What the 58/58 result actually means

All 58 logged events satisfied the implemented phase, amplitude and staging rules in this example. It demonstrates internally consistent event decisions under those settings; the retrospective calibration results on the earlier page are the separate assessment of phase-estimation bias and spread.

DEVELOPMENT / ORIGINAL WORKSPACE

How the project evolved: Days 1–11

The original development work included exploratory scripts and experiments beyond the curated reviewer repository.

STAGE	MAIN QUESTION AND WORK	REPRESENTATIVE ORIGINAL WORK
Days 1–3	Can EEG signals and hypnogram stages be aligned, filtered and inspected reliably?	Data inspection, stage mapping, preprocessing previews
Day 4	Can a simple seven-feature model classify stages?	train_eval_stager_sklearn.py; LR baseline
Day 5	Can predictions update as EEG arrives in chunks?	run_stager_streaming.py, 30-s ring buffer
Days 6–7	Can SO phase candidates be detected and gated by NREM?	so_phase_preview.py, candidate and trigger plots
Day 8	How do one- and two-click trains behave?	run_closed_loop_clicks_sim.py; logs
Days 8.5–9	Are click phases near the retrospectively evaluated target?	day9C_calibrate_target_phase.py; phase error
Day 9D	Would per-record calibration transfer to a later segment?	day9D_subject_calib_then_eval.py
Day 10	Which stage models and gates work in the controller?	day10_make_dataset.py, day10_train_models.py, stress tests
Day 11	Could EEGNet replace handcrafted features?	CNN architecture and exploratory integration

Three decisions that shaped the implementation

Confident Wake→NREM errors led to channel-agreement gating. Phase-at-click diagnostics revealed that the sign of the phase correction mattered. Separating slow stage updates from fast SO timing required causal filters, timestamp ordering and a stateful scheduler.

This is a functional development map, not a list of every temporary source file. Some original experimental scripts are not included in the compact reviewer repository.

CURRENT REVIEWER REPOSITORY

Code map for the reviewer

Where to find the current implementation, and how the components relate to the figures.

PATH IN CURATED REPOSITORY	WHAT TO INSPECT
src/sleepcl/staging/features.py	Welch bandpowers, standard deviation and successive difference
scripts/day10_make_dataset.py	EDF/hypnogram pairing, valid epochs and feature cache
scripts/day10_train_models.py	LR/SVM/RF/MLP fit, participant-separated validation and model bundles
src/sleepcl/stream.py; preproc.py	Chunk replay, buffers and causal filtering
src/sleepcl/so_phase/detector.py	Moving-window Hilbert phase, amplitude and candidate rule
scripts/run_closed_loop_clicks_sim.py	End-to-end replay, stage-confidence gates, --ch2 and armed scheduler
scripts/day10_stress_test_gating.py	Selected-window controller comparisons and aggregation
src/sleepcl/staging/torch_eegnet.py	Exploratory EEGNet-like model definition

What a reviewer can conclude now

The code implements feature-based staging, online-style rolling prediction, SO phase candidates and a rule-based simulated-click controller. In four participant-separated development folds, the MLP has the highest average classical staging scores, but Wake and N1 remain difficult. The representative LR replay shows that a controller can behave consistently while failing to distinguish N2 from N3.

What is still worth testing

Larger participant-separated evaluation; better probability calibration; a matched comparison of frontal-only versus frontal/posterior gating; and a prospective check of the chosen SO phase reference and any actual audio-chain delay.

Bottom line

The interesting result is not just that clicks can be scheduled: it is how staging errors, phase conventions, channel information and timing rules interact to determine when they occur.